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STUDIES OF $^{63/65}$Cu-NQR AND ELECTRICAL RESISTIVITY FOR THE WIDE RANGE OF x IN $La_{2-x}Sr_xCuO_4$

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The nuclear spin-lattice relaxation rate, $1/T_1$, of Cu-NQR in $La_{2-x}Sr_xCuO_4$ is suppressed by superconductivity without the BCS type enhancement just below T_c, while its temperature dependence obeys the Korringa relation below 100 K in the normal region. The clear changes of the enhancement of $1/T_1$ and $1/T_2$ and also of the profile of the Cu-NQR spectra are observed with increasing x beyond $x=0.3\sim0.4$, suggesting the change of the electronic state of the La-system around $x=0.3\sim0.4$. The anomalies of relaxation and intensity of Cu-NQR near $x=1$, where the electronic state is again localized, seems to be newly suggestive of magnetic order.

1. Introduction

While many families differing in structure and constituent elements are discovered in copper based oxide superconductors, they all share Cu–O planes as two dimensional substructures. The one striking feature is that antiferromagnetism is commonly situated near the superconducting phase in both n-(electron-doped) and p- (hole-doped) type oxides [1,2]. It is therefore very important to understand the strong antiferromagnetic spin fluctuations of CuO_2 sheets for high-T_c superconductivity. Although the $La_{2-x}M_xCuO_4$ (M is divalent Sr, Ba and Ca) has the simplest structure, this system involves interestingly rich physical properties such as magnetic, structural and metal–insulator transition as a function of hole-doping [3]. Among various kinds of fundamental measurements, nuclear magnetic resonances are performed successfully to study local electronic and magnetic properties of this system [4].

The divergence of the nuclear spin relaxation rates of ^{139}La-NQR indicates the existence of a peculiar phase between the 3D-antiferromagnetic ($x<0.02$) and superconducting phase ($x>0.06$) at low temperatures of $T_c^*\sim12$ K [5,6]. This transition is accompanied with a magnetic ordering and also anomalous shifts of the resonance frequency of ^{139}La-NQR to lower frequency side [6]. This fact implies that the magnetic order of Cu moments below T_c^* induces an anomalous change of electric quadrupole interactions. We reported also the observation of Cu-NQR in the superconducting region of the $La_{2-x}Sr_xCuO_4$ system [7]. It has been revealed that nuclear relaxation behavior is similar to that of the superconducting $YBa_2Cu_3O_7$ [8], although the Cu-spectra show rather complex profiles for $x<0.3$, the origin of which is still unresolved [9,10].

In order to clarify the effect of the doped-holes on characteristic spin and charge fluctuations under the strong electron correlations in the CuO_2 sheets, we have intensively investigated Cu- and La-NQR of the La_2CuO_4 system in a wide range of Sr-substitution. In this paper, we report the details of the temperature and concentration dependences of Cu-nuclear relaxation in the wide range of x up to $x=1$ of the $La_{2-x}Sr_xCuO_4$. We also report the anomalies of the Cu-NQR, which suggest the existence of a new antiferromagnetic phase near $x=1$ where the system becomes again insulating. The extended results of the ^{139}La-NQR in a small hole-doped region will be reported separately [11].

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2. Experimental

All samples were prepared by solid state reaction from mixed powder of starting materials of La_2O_3, $BaCO_3$, $SrCO_3$ and CuO with 99.99% purity. The detailed conditions for the heat treatments were reported in the previous papers [5,6]. The reacted pellets were crushed into 350-mesh powder for NMR measurement. The X-ray diffraction analysis indicated that the samples up to $x=1$ crystallized in a single phase with the K_2NiF_4 type structure. The lattice constant of the present samples is shown as a function of Sr-concentration in fig. 1. The lattice constant of the c-axis increases with increasing x up to $x\sim 0.3$ and then decreases rapidly for $x>0.4$. The superconducting transition temperatures were obtained by the resistivity and magnetic susceptibility measurements [12]. The bulk superconductivity was confirmed in the range of $0.08\leqq x\leqq 0.25$ by the heat capacity jump at T_c [13] as well as by the magnetic susceptibility measurement. The electrical resistivity was measured by the DC four probe method and indium soldered contacts. A conventional phase coherent pulse NMR apparatus was used for measurement of the relaxation and NQR spectra of La and Cu nuclei. The nuclear spin-lattice relaxation time was measured by the saturating comb pulse method.

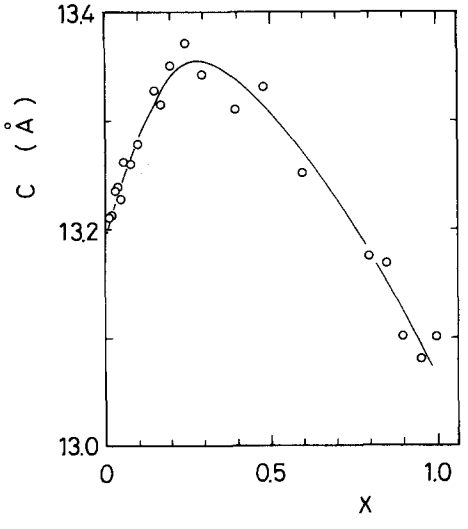


Fig. 1. Lattice parameter of c-axis as a function of x in $La_{2-x}Sr_xCuO_4$ obtained by the X-ray measurement at room temperature.

3. Results

3.1. Electrical resistivity

Fig. 2 shows the temperature dependence of the electrical resistivity for various compositions of $La_{2-x}Sr_xCuO_4$. As is well known, the undoped La_2CuO_4 is an insulating antiferromagnet of $T_N\sim 270$ K, and the system remains insulating for small hole-doping (Sr concentration) up to $x=0.05$. The metal–insulator transition takes place near $x=0.05$, and superconductivity appears in a suitable hole concentration between $0.06<x<0.25$ where the normal state resistivity above T_c changes linearly with temperature, showing that the system is metallic.

We confirmed that the samples were crystallized in a single phase up to $X=1$ by the x-ray diffraction measurement. The result is consistent with the previous one that the K_2NiF_4 type compound is synthesized in the large composition range $0\leqq x\leqq 1.34$ [14]. The measurement of the temperature dependence of electrical resistivity shows semiconducting behavior for the more concentrated region of $x\geqq 0.8$ as shown in fig. 2. The concentration dependence of the resistivity at 300 K is shown in fig. 3. The resistivity shows minimum near $x=0.3$–0.4, and increases with increasing x. The resistivity for $x=1$ is larger than that of the insulating La_2CuO_4. As shown in fig. 4, the temperature dependence of resistivity for $x\geqq 0.8$ is apparently fitted by the relation of log

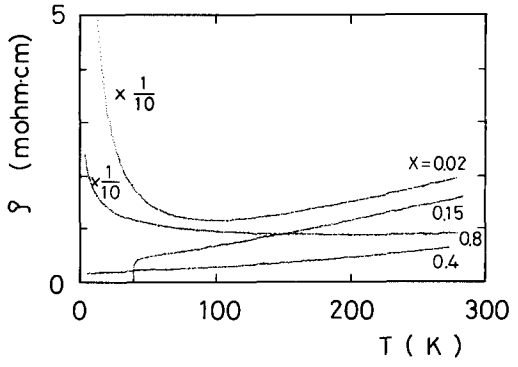


Fig. 2. Temperature dependence of the electrical resistivity for various x in $La_{2-x}Sr_xCuO_4$.

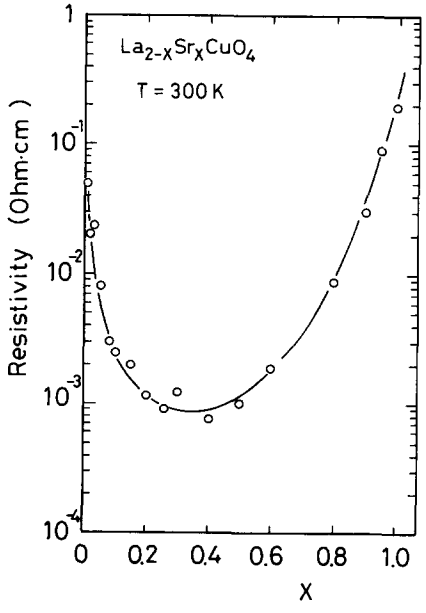


Fig. 3. Concentration dependence of the electrical resistivity at room temperature in $La_{2-x}Sr_xCuO_4$.

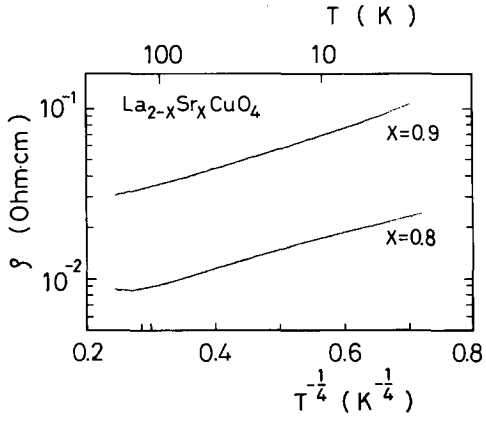


Fig. 4. Electrical resistivity as a function of $T^{-1/4}$ in $La_{2-x}Sr_xCuO_4$ of $x=0.8$ and 0.9.

$\rho \propto \exp(T^{-1/4})$ below 300 K, suggesting that the variable-range hopping conduction [15] is dominant for $x \geqq 0.8$, and that the electronic state is localized near the Fermi energy.

3.2. $^{63/65}$Cu-NQR

The spin echo signals of $^{63/65}$Cu are observed above 30 MHz in zero external field for $x \geqq 0.12$. The failure of the observation of the NQR signals for $x \leqq 0.1$ is caused by their very shortened spin–spin relaxation time [7]. Fig. 5 shows the zero field spectra of $^{63/65}$Cu at 1.4 K for various concentrations. As reported previously [7], the Cu-spectra for $x \leqq 0.3$ consist of three peaks, indicating the existence of another set of Cu signals, while the spectra for $x \geqq 0.4$ show only two peaks corresponding to two isotopes of $^{63/65}$Cu as shown in fig. 5. The line profiles are not modified by various conditions for sample preparation. The quite similar spectra are also obtained for Ba-doping.

The nuclear spin-lattice relaxation rate, $1/T_1$, was measured at each peak in the spectra. The recovery curve of the nuclear magnetization after the saturation is almost single exponential around $x=0.4$ from 1.5 to 100 K. However, non-single exponential behavior appears for small x ($x \leqq 0.3$) and for large x ($x \geqq 0.6$). Such non-single exponential behavior in the superconducting samples becomes significant with decreasing temperature. The recovery curves seem to be described in some cases by the root exponential relation, indicating that values of T_1 are

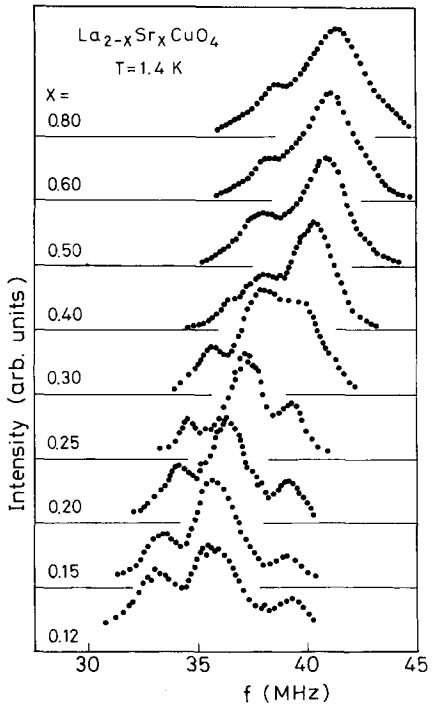


Fig. 5. Spin echo spectra of $^{63/65}$Cu in $La_{2-x}Sr_xCuO_4$ at $T=1.4$ K.

distributed largely. The profile of the recovery curve does not depend on the saturating condition. Therefore, the distribution in T_1 is not due to the spin diffusion process. As the each component of the magnetization recovery curve changes similarly against temperature, the temperature dependence of T_1 is considered to change similarly among each component. Thus, we show, for simplicity, the temperature dependence of the longest component of $1/T_1$, which is extracted from a tail part of the recovery curve.

In the superconducting samples of $x \leqq 0.25$, $1/T_1$ below T_c is much suppressed as is shown in fig. 6. The temperature dependence of $1/T_1$ is very similar to that of Cu in the CuO_2 plane of the Y-system [8], namely, $1/T_1$ is suppressed by superconductivity without the BCS type enhancement just below T_c, and decreases with power-law-like behavior with decreasing temperature. This observation indicates that the spin fluctuations of Cu moments are suppressed by the superconducting energy gap. $1/T_1$ is much enhanced and is almost at the order of ms^{-1} in the normal state for $x=0.2$. Although the opening of a gap of spin excitation spectrum above T_c is suggested by the neutron scattering experiment on the single crystal [16], the present T_1 measurement below 100 K does not make clear this point. It is perhaps necessary to measure the temperature dependence of $1/T_1$ up to high temperatures of 300 K in the La-system [17].

For the range of $x \geqq 0.3$ where the electrical resistivity shows metallic behavior, the temperature dependence of $1/T_1$ obeys the Korringa relation, $T_1T=$const., between 1.5 to 100 K as shown in fig. 7. (For $x \geqq 0.6$, the deviation from the Korringa relation becomes dominant below 30 K. (See also fig. 9.) In this case, we deduce simply the T_1T values from the narrow range of high temperatures.) This means the system is metallic in local nature and that the thermal excitations of conduction electrons at the Fermi level are sizeable at the Cu sites. Fig. 8 shows the x-dependence of the $(T_1T)^{-1}$ in the normal state. $(T_1T)^{-1}$ deceases dramatically with x up to $x=0.3$ and attains the constant value of $(T_1T)^{-1}=2.0\ (\mathrm{sK})^{-1}$ for $x \geqq 0.4$. Moreover, the independence of T_1 on x for $x \geqq 0.4$ can be verified by the fact that the data points at 77 K for $0.5 \leqq x \leqq 0.8$ are distributed on the same recovery curves within the experimental scattering, showing that both the

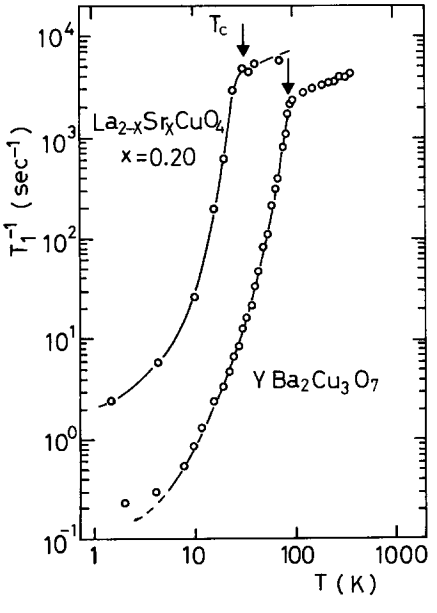


Fig. 6. Temperature dependence of the nuclear spin–spin relaxation rate, $1/T_1$, of ^{63}Cu in the superconducting $La_{1.8}Sr_{0.2}CuO_4$. The reported result (from ref. [8]) in $YBa_2Cu_3O_7$ is also shown.

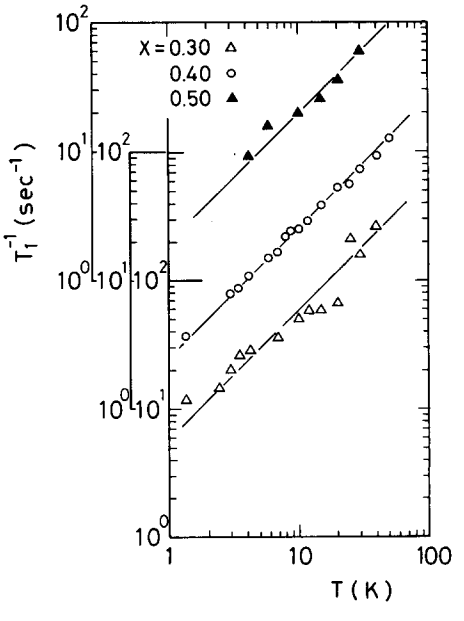


Fig. 7. Temperature dependence of the nuclear spin–spin relaxation rate, $1/T_1$, of ^{63}Cu in the normal metallic samples of $La_{2-x}Sr_xCuO_4$.

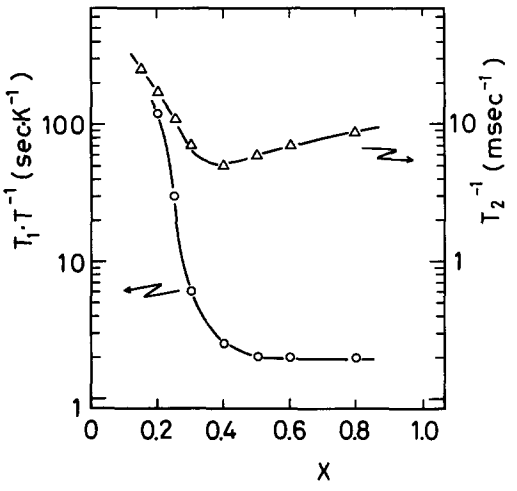


Fig. 8. Concentration dependence of $(T_1T)^{-1}$ (○) and $1/T_2$ (△) of ^{63}Cu in $La_{2-x}Sr_xCuO_4$.

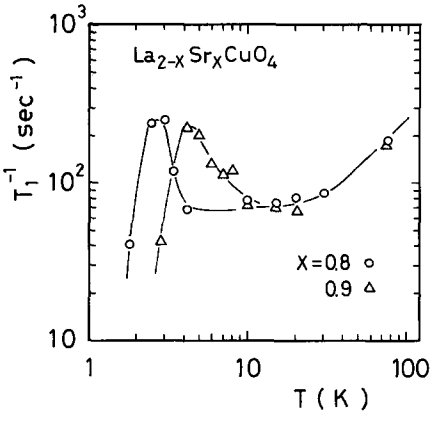


Fig. 9. Temperature dependence of the nuclear spin-lattice relaxation rate, $1/T_1$, of ^{63}Cu in the semiconducting samples of $x=0.8$ and 0.9 in $La_{2-x}Sr_xCuO_4$.

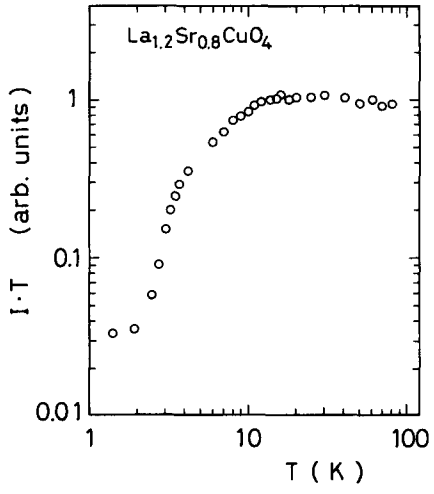


Fig. 10. Intensity of the Cu-NQR times temperature, IT, as a function of temperature for $x=0.8$.

short and long components of T_1 are essentially independent of x for $x \geqq 0.4$.

In addition to the remarkable change of T_1, the spin–spin relaxation rate, $1/T_2$ at 1.4 K decreases rapidly with increasing x up to $x=0.3$ as shown in fig. 8. For $x \geqq 0.4$, $1/T_2$ increases slightly with x. The spin–spin relaxation time is usually determined by dipolar coupling between Cu nuclear spins and is expected to remain nearly constant in the order of ms because of the unchanged nuclear number and the space distance between Cu atoms for Sr-doping. Therefore, the extremely shortened T_2 is attributed to another origin such as the indirect interactions of Cu nuclei through magnetic Cu moments. Thus, it should be emphasized that the clear difference of the x-dependence of nuclear relaxation times, T_1 and T_2, must be related with the change of the electronic state beyond $x=0.3$–0.4 in the La-system.

For the more concentrated region $x \geqq 0.8$ where the system is semiconducting (insulating) as seen in figs. 2 and 3, the anomalies in the temperature dependence of the $1/T_1$ appear at low temperatures as shown in fig. 9. The temperature at which $1/T_1$ shows a peak in its temperature dependence increases with increasing x. In addition to the relaxation anomalies, the intensity of the spectra decreases significantly near the peak temperature in relaxation as shown in fig. 10. Moreover, the Cu signals for $x=0.95$ and 1 are almost wiped out from the range of around 35–45 MHz at least below 77 K. These experimental facts seem to suggest that a kind of the magnetic order (possibly antiferromagnetic one) again appears for $x \geqq 0.8$ similar to that in the undoped and in the dilutely-doped region where the system is insulating (semiconducting). The magnetic ordering temperatures are 2.5 and 4 K for $x=0.8$ and $x=0.9$, respectively, and are expected to be far above 77 K for $x \geqq 0.95$.

4. Discussion

Our observation of the split spectra of the Cu-NQR for $x<0.4$ [7] was experimentally confirmed by the following studies [9,10]. Two possibilities can be considered to interpret the split NQR spectra: i.e., magnetic origin and electric quadrupolar one. We suggested previously that the splitting of the Cu-NQR lines is owing to hyperfine fields originated from long-range magnetic order of Cu moments. This suggestion seemed to be consistent with the results of μSR [18] and heat capacity [13] measurements. On the contrary, the recent NMR measurements claim that the complex spectra are attributed to the distribution of the electric quadrupole interaction due to the oxygen vacancies and/or due to the random distribution of Sr and La around Cu [9,10]. The effect of the charge differentiation on the EFG is proposed similar to that in the Y-system [10].

Until now after our first report [7], the detailed measurements on the temperature dependence of T_1 for $0.12 \leqq x \leqq 0.4$ have not shown any anomalies below 77 K, which is not suggestive of the existence of apparent long-range magnetic order. Including the NMR results [9,10], any other nuclear magnetic resonance investigations do not disclose clear evidence for the existence of static magnetic Cu-moments down to 1.4 K for $x<0.3$–0.4, although the origin of the separated spectra for $x<0.4$ is not fully resolved at the present time. Therefore, our previous interpretation based on the long-range magnetic order with reduced magnetic Cu moments seems to be unlike that for the origin of the separated spectra.

It should be noted, however, that the complex spectra with more than two peaks appear only for $x<0.4$ where the remarkable enhancement of $1/T_1$ and $1/T_2$ are observed. In the La-system, T_2 increases from 20 μs for $x=0.12$ to 200 μs for $x=0.3$ as shown in fig. 8. In the previous paper, we suggest that the extremely shortened T_2 is attributed to the indirect nuclear coupling via virtual antiferromagnetic spin excitations [7]. Interestingly, Pennington et al. [19] show that the T_2 of Cu(2) in $YBa_2Cu_3O_7$ can be explained by a strong indirect coupling between Cu(2) nuclear spins through the Cu moments which are coupled through superexchange interactions to each other. Thus, the large enhancement of $1/T_2$ and $1/T_1$ in the La-system is reasonably considered to be evidence that Cu nuclear spins couple with each other through the electronic Cu moments and their spin fluctuations, and the rapid decreases of T_1 and T_2 with increasing x suggest the suppression of the antiferromagnetic fluctuations by hole-doping. The suppressed relaxation rates for $x>0.4$ can be considered to result from the diminution of the effective magnetic moments by hole-doping. A possible interpretation seems to be that the doped hole spins on the oxygen ions form nonmagnetic singlet states with adjacent Cu^{2+} spins which have no magnetic interaction with other d-holes [20]. In this case, the anomalous splitting of the NQR spectra as well as the enhancement of nuclear relaxation rates may not be expected any more for the highly hole-doped region.

The quantities such as susceptibility, Hall coefficient and thermoelectric power change significantly at near $x=0.3$ [21], implying that the electronic state in the normal state is much altered by hole-doping near $x=0.3$. The decrease of T_c for $x>0.2$ seems to occur concurrently with the change of the normal state quantities. Thus, one may safely say that the Cu-NQR of the La-system provides another evidence of the anomalous change of the magnetic state of Cu spins for hole-doping around $x=0.3$–0.4.

5. Summary

We have investigated the Cu-NQR and electrical resistivity in the wide range of Sr-substitution. The lattice constant changes remarkably with x. The concentration dependence of the lattice constant seems to anti-correlate with that of the electrical resistivity, i.e., the increase of the lattice constant of the c-axis is accompanied by the decrease of electrical resistivity for Sr-substitution. The La-system undergoes the insulator-to-metal transition near $x=0.05$ and again the metal-to-insulator transition for $x \geqq 0.8$. In the region where the electronic state is localized near $x=1$, the loss of Cu-NQR intensity with decreasing temperature is accompanied by the anomalous enhancement of $1/T_1$. The magnetic order (possibly antiferromagnetic order) seems to be newly disclosed in the insulating region for $x \geqq 0.8$.

In the superconducting samples of $x \leqq 0.25$, $1/T_1$ is suppressed without the BCS type enhancement just

below T_c, and decreases with power-law-like behavior with decreasing temperature. In the normal state for $0.3 \leqq x \leqq 0.6$, the Korringa relation, $T_1T = \text{const.}$, is observed below 100 K. Both T_1T and T_2 depend strongly on Sr-concentration for $x \leqq 0.3$. The rapid decreases of $1/T_1$ and $1/T_2$ are concurrently observed with the decrease of T_c, suggesting that the antiferromagnetic fluctuations are correlated with the high-T_c superconductivity. The complex spectra and the large enhancement of the nuclear relaxation rates for $x < 0.4$ are different qualitatively from those for $x > 0.4$, which is consistent with the recent observation of the change of the normal state properties around $x = 0.3$–0.4.

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